

Frontiers of Visual Place Recognition in SLAM

Mohammed Ehab

Abstract—The problem of Visual Place Recognition has gathered a lot of attention in the computer vision community, with very successful recent advances based on deep learning backbones. However, these advances were never tested as part of a SLAM pipeline on real robotics datasets. In this report, we benchmark SOTA VPR models, demonstrate their functionality on indoors robotics datasets without the need for further finetuning, and discuss their integration into the Kimera SLAM system.

I. INTRODUCTION

Visual Place Recognition (VPR) is a crucial sub-problem in many SLAM pipelines. In the context of SLAM, Bag-of-Words [1] has been established as the main approach for this task due to its speed and lack of need for a GPU. However, the computer vision community developed learning-based approaches for this task with higher precision and recall ([2],[3], [4]). With the advent of TinyML, it will become possible for these approaches to run in real time and be relevant for SLAM, raising the need for an integration and evaluation on SLAM datasets.

VPR is naturally framed as a contrastive learning problem, where a neural network learns to map images of places into vectors with nearby vectors corresponding to similar-looking places. NetVLAD [2] is a fundamental paper that employs this contrastive learning approach, while also drawing inspiration from earlier work on VPR by designing an aggregation layer that aggregates local features into a global descriptor. More recent approaches develop on NetVLAD by leveraging recent advances in deep learning as well as designing better aggregation techniques.

II. VIEWPOINT ROBUSTNESS BENCHMARK

As of this report, the most prominent approaches for VPR in the computer vision community are SALAD with Clique Mining [5] and EigenPlaces [3]. SALAD succeeds EigenPlaces and massively outperforms it on all the benchmarks the authors of SALAD use. However, all of the datasets in that benchmark come from a frontal view camera mounted on a car. EigenPlaces incorporates viewpoint robustness directly in its training procedure, and the authors found that it outperforms all the previous models on **multiview** datasets.

Since viewpoint robustness is important for robotics, we carried a benchmark of SALAD against EigenPlaces on the AmsterTime multiview dataset [6] using VPR-Bench [7]. The results are summarized in Table I.

We concluded that SALAD overall outperforms EigenPlaces, and all the evaluation forward should be performed using SALAD.

	Recall @ 1	Recall @ 5	Recall @ 10	Recall @ 20
EigenPlaces	48.9	69.5	76.0	81.4
SALAD	60.9	82.1	86.4	89.8

TABLE I: EigenPlaces vs SALAD on AmsterTime.

III. SALAD’S IMPROVEMENTS ON NETVLAD

In this section, we summarize the improvements SALAD makes over NetVLAD to get its superior performance.

- **Architecture:** SALAD uses the more recent ViT [8] architecture as opposed to ResNet in NetVLAD.
- **Finetuning:** NetVLAD is trained from scratch while SALAD is finetuned from the large foundation model DINOv2.
- **Aggregation:** NetVLAD learns cluster centers for the local features and assigns features to clusters via a softmax. SALAD learns an abstract score matrix that assigns features to clusters and uses Sinkhorn’s optimal transport algorithm [9] for aggregation. This procedure has also shown success in feature matching with Super-Glue [10].
- **Loss Function:** NetVLAD uses the triplet loss function that compares an anchor with a single positive and a negative example. SALAD uses the multi-similarity loss [11] that can handle multiple positive and negative examples.
- **Descriptor Size:** SALAD’s descriptor is 4 times smaller than NetVLAD’s descriptor, which is great for robotics applications. They are able to reduce dimensionality by showing a procedure that lets the model learn to discard uninformative features coming from DINOv2.

Furthermore, in another paper, the authors of SALAD proposed an improvement in how batches are mined during training [5]. They show a method called Clique Mining to obtain difficult negative examples and show that by using a batch consisting of half difficult examples and half easy examples, the model’s performance can be significantly improved.

IV. INDOORS RECALL DEMO

To test SALAD’s recall in an indoor setting, we ran an experiment on the TUM-RGBD SLAM dataset [12], where frames are only added to the database 5 seconds after they take place. In this video demonstration, A scene with a close-up on a chair is shown, and the model recalls the closest scene it saw where the chair is in the background. After enough time passes, a frame with the chair up close is added to the database, and the model immediately switches to that scene. Furthermore, after the model sees scenes from all over

the room, it is essentially capable of tracking the live video using scenes it recalls. The code for this experiment can be found here.

V. INTEGRATION INTO KIMERA

To integrate into Kimera [13], we wrote a script that traces SALAD into TorchScript, a format that can be read by the C++ distribution of PyTorch. We use FAISS [14] to perform K-nearest-neighbors queries efficiently. The code and details on how to run can be found here.

VI. EVALUATION

By default, Kimera-VIO runs on the Euroc MAV dataset [15]. We ran an evaluation of BoW vs SALAD on the MH_01_easy and the MH_04_difficult sequences.

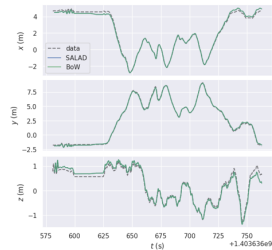


Fig. 1: MH_01_easy

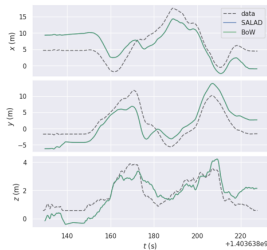


Fig. 2: MH_04_difficult

We did not see any differences in the estimate provided by BoW compared to SALAD. Investigating the Euroc dataset, we see that it does not test loop closures at all. Rather, in the more difficult sequences, the MAV moves and accelerates faster.

A last-minute evaluation was attempted on the ADVIO dataset [16]. The dataset offers more opportunity to test loop closures, with ground-truth paths akin to the ones in Figure 3. The evaluation was attempted by converting ADVIO to the Euroc format using this script. However, even the original Kimera fails to run on the output dataset, indicating that there is still a formatting issue.

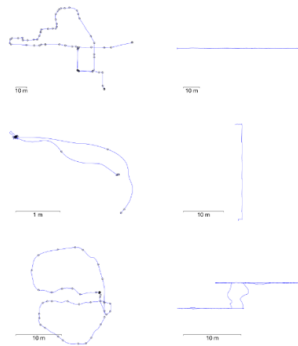


Fig. 3: Sample ADVIO Paths

VII. CONCLUSION

SALAD with Clique Mining is a very promising improvement on SLAM pipelines, and is currently the SOTA in VPR on all accords. It can work indoors without further finetuning.

A working Kimera integration of SALAD is provided but needs further evaluation on a fair dataset. Future work includes performing this evaluation as well as measuring the gains from finetuning SALAD for SLAM-specific datasets.

VIII. REFERENCES

REFERENCES

- [1] D. Galvez-López and J. D. Tardos, “Bags of Binary Words for Fast Place Recognition in Image Sequences,” *IEEE Transactions on Robotics*, vol. 28, no. 5, pp. 1188–1197, Oct. 2012. [Online]. Available: <http://ieeexplore.ieee.org/document/6202705/>
- [2] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, “NetVLAD: CNN architecture for weakly supervised place recognition,” May 2016, arXiv:1511.07247 [cs]. [Online]. Available: <http://arxiv.org/abs/1511.07247>
- [3] G. Berton, G. Trivigno, B. Caputo, and C. Masone, “EigenPlaces: Training Viewpoint Robust Models for Visual Place Recognition,” Aug. 2023, arXiv:2308.10832 [cs]. [Online]. Available: <http://arxiv.org/abs/2308.10832>
- [4] S. Izquierdo and J. Civera, “Optimal Transport Aggregation for Visual Place Recognition,” Jun. 2024, arXiv:2311.15937 [cs]. [Online]. Available: <http://arxiv.org/abs/2311.15937>
- [5] —, “Close, But Not There: Boosting Geographic Distance Sensitivity in Visual Place Recognition,” Jul. 2024, arXiv:2407.02422 [cs]. [Online]. Available: <http://arxiv.org/abs/2407.02422>
- [6] B. Yildiz, S. Khademi, R. M. Siebes, and J. v. Gemert, “AmsterTime: A Visual Place Recognition Benchmark Dataset for Severe Domain Shift,” Jun. 2022, arXiv:2203.16291 [cs]. [Online]. Available: <http://arxiv.org/abs/2203.16291>
- [7] M. Zaffar, S. Garg, M. Milford, J. Kooij, D. Flynn, K. McDonald-Maier, and S. Ehsan, “Vpr-bench: An open-source visual place recognition evaluation framework with quantifiable viewpoint and appearance change,” *International Journal of Computer Vision*, pp. 1–39, 2021.
- [8] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” Jun. 2021, arXiv:2010.11929 [cs]. [Online]. Available: <http://arxiv.org/abs/2010.11929>
- [9] M. Cuturi, “Sinkhorn Distances: Lightspeed Computation of Optimal Transport,” in *Advances in Neural Information Processing Systems*, vol. 26. Curran Associates, Inc., 2013. [Online]. Available: https://papers.nips.cc/paper_files/paper/2013/hash/af21d0c97db2e27e13572cbf59eb343d-Abstract.html
- [10] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, “SuperGlue: Learning Feature Matching with Graph Neural Networks,” Mar. 2020, arXiv:1911.11763 [cs]. [Online]. Available: <http://arxiv.org/abs/1911.11763>
- [11] X. Wang, X. Han, W. Huang, D. Dong, and M. R. Scott, “Multi-Similarity Loss with General Pair Weighting for Deep Metric Learning,” Mar. 2020, arXiv:1904.06627 [cs]. [Online]. Available: <http://arxiv.org/abs/1904.06627>
- [12] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A benchmark for the evaluation of RGB-D SLAM systems,” in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Oct. 2012, pp. 573–580, ISSN: 2153-0866. [Online]. Available: <https://ieeexplore.ieee.org/document/6385773>
- [13] A. Rosinol, M. Abate, Y. Chang, and L. Carlone, “Kimera: an Open-Source Library for Real-Time Metric-Semantic Localization and Mapping,” Mar. 2020, arXiv:1910.02490 [cs]. [Online]. Available: <http://arxiv.org/abs/1910.02490>
- [14] M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P.-E. Mazaré, M. Lomeli, L. Hosseini, and H. Jégou, “The faiss library,” 2024.
- [15] “The EuRoC micro aerial vehicle datasets - Michael Burri, Janosch Nikolic, Pascal Gohl, Thomas Schneider, Joern Rehder, Sammy Omari, Markus W Achtelik, Roland Siegwart, 2016.” [Online]. Available: <https://journals.sagepub.com/doi/full/10.1177/0278364915620033>
- [16] S. Cortés, A. Solin, E. Rahtu, and J. Kannala, “ADVIO: An authentic dataset for visual-inertial odometry,” Jul. 2018, arXiv:1807.09828 [cs]. [Online]. Available: <http://arxiv.org/abs/1807.09828>